Nathaniel Recto

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Education

Bachelor of Engineering, Computer Engineering Sep 2022 – Jun 2026
Toronto Metropolitan University *Toronto, ON*

Technical Skills

Computer Vision: Object detection (YOLOv8), image classification (MobileNetV2), object tracking, transfer learning, OpenCV, MediaPipe.
Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, NumPy, Pandas, Matplotlib, fine-tuning, hyperparameter tuning.
Experiments and Statistics: Experiment design, leakage-safe dev/test splits, cross-validation, confidence intervals, data labelling.
Model Reliability: Probability calibration, threshold tuning, failure-mode analysis, mAP50, ROC-AUC, F2, Brier score.
Deployment and Software: Core ML, TensorFlow Lite, on-device inference, SwiftUI, Flask, SQLite, Git/GitHub, Linux, automated testing.
Languages: Python, C/C++, Java, Swift, Kotlin, SQL.

Projects

Basketball Shot Tracker (iPhone App) | *Python, PyTorch, YOLOv8, OpenCV, NumPy, Swift* Oct 2026

- Built a computer-vision system that calls basketball shots made/missed from one phone camera, pairing a fine-tuned **YOLOv8s** object detector (**8,347** images, **0.85** val mAP50) with a physics-based tracking algorithm.
- Designed a leakage-free experiment: hand-labelled **555** real-world shots blind from **16** public clips and **11** own sessions in **2** gyms, split dev/test by session before filming, and scored the test set once with frozen, hashed code.
- Achieved **85.6%** end-to-end accuracy (**95%** CI **77–91%**) on **90** held-out shots in an unseen gym (**87** found, **0** false alarms); failure analysis guided tracker and detector fixes lifting own-footage accuracy **13.4%** to **78.2%**.
- Deployed it as a native **Swift/SwiftUI** iPhone app running **Core ML** on the Neural Engine at **28–32** fps (**45–62** ms per call), verified by **132** automated tests to match **Python**'s **123** shot calls exactly across **33,386** frames.

Real-Time Driver Drowsiness Detection System | *Kotlin, Python, TFLite, MediaPipe* Sep 2025 – Apr 2026

- Built a driver-safety Kotlin Android app running 2 TensorFlow Lite CNNs with CameraX and MediaPipe, detecting drowsiness in real time via on-device inference.
- Curated a 4-class eye and mouth image dataset, then fine-tuned 2 MobileNetV2 CNN classifiers via transfer learning, achieving **99.17%** eye-state accuracy and **96.18%** yawn-detection accuracy.
- Designed a scoring algorithm fusing 4 drowsiness signals (eye closure, yawns, head tilt, and model output) that triggers driver warnings once the combined score reaches the 100-point threshold.
- Evaluated model outputs across lighting, head angles, facial movements, and combined drowsiness scenarios, identifying failure modes and defining operational limits for on-device inference.

Heart Disease Risk Prediction System | *Python, scikit-learn, XGBoost, Pandas, Flask, SQLite* Oct 2026

- Built an ML pipeline for heart disease screening on **918** UCI patient records from **4** hospitals, using a stratified **80/20** split (**734** train, **184** test) and leakage-safe scikit-learn Pipelines for imputation, scaling, and encoding.
- Evaluated Logistic Regression, Random Forest, and XGBoost using stratified **5**-fold cross-validation and RandomizedSearchCV to optimize F2 for recall, selecting XGBoost with the strongest cross-validated F2 of **0.85**.
- Improved model reliability by calibrating probabilities and tuning thresholds on out-of-fold predictions: **99%** recall (**101** of **102** sick patients), **0.90** ROC-AUC, and a **0.125** Brier score on the test set, cutting missed cases **13** to **1**.
- Served the models through a Flask web app with a **4**-step assessment, **3** selectable models, per-prediction explanations, and private SQLite/SQLAlchemy history, validated by **7** automated tests.

Experience

Operations Support May 2024 – Aug 2024, May 2025 – Aug 2025
Enercare *Markham, ON*

- Processed contractor payments, invoices, and work orders in GERS, Clarify, and MS Office, reducing processing time by 20% through accurate record updates.
- Coordinated administrative requests and escalations as the key team contact, using tracked follow-ups to increase business-support productivity by 15%.